

Teach the evidence chain

Companion to the 50-page CNC Plasma Cutter student workbook. It contains the correct option and rationale for all 150 source questions, guidance for every applied task and folio stage, and illustrative responses to the 15 original extended prompts.

CURRENT ACCESS BOUNDARY Students work on design, offline file preparation and evidence interpretation. Actual referencing, jogging, sensing, firing, physical simulation and cutting remain with trained, authorised teachers under the approved school procedure.

A teachable rhythm

Present the linked lesson and selected figures, read the paper synthesis, ask students to explain the idea, then use the original check questions for retrieval. On the third page, the applied activity or folio record supplies evidence for the supported response. Students may use the table and ruled area together; do not require a second redundant essay.

Questions use the source order and original choices. Discuss the reasoning, then invite a visible correction. They are formative; this guide does not invent a mark allocation, pass threshold, due date or operator sign-off.

Flexible delivery

The source proposes an optional Year 10 ten-week pathway. Confirm the school programme, lesson lengths and student needs before scheduling. All lessons can use the source-supported paper or offline pathway when live demonstrations are unavailable. State clearly when a physical activity did not occur.

Using the teacher pages

Each lesson has one answer-key page followed by one activity/response guide. Accept equivalent source-supported explanations, especially for open design tasks. A sample response demonstrates reasoning and must not be represented as an actual student result.

Student pages 48–50 supply a coordinate reference, enlarged evidence fields and key words/sources. Guidance for these pages appears at the end of this teacher guide.

Open teacher resources and the current programme

Confirm before delivery

Authority and practical preparation

The source teacher programme remains provisional. Faculty sign-off, delivery year, timetable and current local operating arrangements require confirmation. Its school safety citation establishes teacher-only machine use; supplier illustrations explain the system and do not approve the installed equipment.

Confirm and record

Local reference / decision

Trained authorised operator and required school approval

Current risk assessment, operating procedure and supervision

Extraction, screening, guarding, controlled access and operator separation

Electrical isolation, cooling, retrieval and safe inspection

Installed equipment, consumables, material and verified source chart

CAD/CAM applications and verified post-processor responsibility

Final student brief, tolerances, formal assessment and dates

A KEY DISTINCTION The website setup trainer is screen-only and has no machine connection. SureControl Simul followed by Start moves the actual machine. They must never be described as equivalent student practice.

Students are not required to create an executable TAP file in this sequence. They can explain the CAD → DXF → CAM → compatible program → control pathway and supply a traceable handover for teacher review. A teacher must verify process configuration and executable output.

Use the source teacher-start and guided operating reference with the authorised operator. This workbook is a learning companion, not a substitute operating procedure.

How a drawing becomes metal

01 B — What directly describes the plasma cutting process?

An electrical plasma arc melts a narrow path and gas removes molten metal. The arc supplies concentrated heat and gas flow clears the molten material.

02 A — What does kerf mean here?

The gap produced along the cut. Kerf is the width of material removed along the cutting path.

03 C — Which component supplies electrical energy for the plasma process?

The plasma power source. The power source provides the electrical energy used in the cutting process.

04 B — Why does extraction belong in the system explanation?

It manages fumes under the approved installation arrangements. The required controls are part of the system even when they do not change the drawing.

05 C — Which record is the best starting point for a later design change?

The native CAD master. The master preserves the editable design from which a revised exchange file can be exported.

06 A — What is the main role of a DXF in this workflow?

Exchange drawing geometry between applications. DXF is the separate drawing handoff, which still needs receiving checks.

07 C — A file ends in .tap. What can you conclude from the extension alone?

It has the supplier workflow's program extension, but compatibility still needs checking. A TAP extension is useful identification, not proof of a matching post-processor.

08 A — Which stage chiefly organises a cutting path and sequence from checked geometry?

CAM preparation. CAM prepares the manufacturing path and process using the verified setup.

09 B — The hole has the wrong diameter in the original drawing. Where should that error first be corrected?

In the native CAD design, followed by a new checked export. Correcting the source design keeps later files traceable to the intended geometry.

10 C — An official video shows a material being cut. What does that establish for this course?

It demonstrates that example; the local material and process still need teacher approval. A useful demonstration remains evidence about its own setup.

Support the explanation

Match the task to the system

A → CAD; B → CAM; C → CNC control; D → power source; E → extraction; F → table. CAD describes the shape, CAM prepares production choices, and control executes prepared instructions. The DXF exchanges geometry; its dimensions, units, contours and revision still need checks, followed by the teacher's verified CAM/post-processor review.

Look for this evidence

- Six correct matches, with reasons explaining two distinctions.
- A handover explanation identifying geometry and preparation/compatibility checks.
- Feedback corrections remain identifiable.

Valid alternatives

Accept different pairs for the explanations when their distinct roles are accurate.

Original extended response

Explain how a student's proposed square-and-hole drawing becomes a teacher-reviewed production job. Identify what CAD, CAM and CNC control each contribute, and justify two checks that prevent a plausible-looking file being mistaken for a ready-to-cut program.

Illustrative response

CAD records the 100 × 100 mm square and central Ø30 mm hole in an editable master. A named DXF revision exchanges its geometry. Checking dimensions and two closed contours after import can reveal a scale or boundary problem. CAM prepares the torch route and sequence; the authorised teacher verifies the process and post-processor before compatible output reaches CNC control. SureControl executes the prepared instructions. A DXF's appearance does not prove its toolpath or compatibility. Students prepare evidence offline; a correct file does not authorise them to operate.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Correctly distinguishes CAD, CAM and CNC control.
- Connects checks to specific geometry, file or process risks.
- Keeps file readiness separate from operator authorisation.

A useful discussion example

The offline teaching drawing is a 100 × 100 mm square with a central Ø30 mm hole. In CAD, confirm the dimensions and save the master. Export a named DXF revision. In the receiving application, check the same size and two intended closed contours. Then the trained teacher reviews the production preparation. A correct circle in CAD does not prove that the transferred file has the same units or that the toolpath is suitable.

Student evidence page 5. Full-size sketch and selected-image fields: student page 49.

Your role in the workshop

01 C — Which is an appropriate student contribution described here?

Prepare a dimensioned drawing on the designated offline computer. Students can contribute actively through approved offline design and preparation.

02 A — Why is watching only for a bright arc an incomplete risk check?

Hot stock, electrical energy and moving parts create other hazards. The process has hazards beyond the visible arc, including heat, electricity and movement.

03 B — How should protective equipment relate to extraction and guarding?

It supports the approved controls rather than replacing them. Different controls address different parts of the risk, so they work together.

04 C — Where must the teacher be during automatic cutting?

At the approved position, physically separated by the screened area and welding curtains. The teacher also requires physical separation during automatic cutting.

05 A — The program has finished. What establishes whether students may enter the shielded area?

The required electrical isolation and the teacher's controlled access decision. Students enter only under the school's isolation and access arrangements.

06 B — How does SureControl Simul differ from an offline CAM preview?

SureControl Simul moves the real machine without firing the torch. That motion is why its operation remains with the trained teacher.

07 C — What is the student response to unexpected movement during observation?

Keep clear and alert the teacher through the agreed arrangements. Reporting from a safe position respects the operating and access boundaries.

08 A — Which evidence supports teacher readiness in this section?

Comprehensive supplier instruction, principal approval and approved site arrangements. These are the stated operating requirements; the course supports but does not replace them.

09 B — Which journal statement is appropriately limited?

I explained the dry-run boundary and recorded the teacher's demonstration. This separates a learning explanation from an actual teacher-led observation.

10 C — Why must the class identify the correct computer for student practice?

An offline design workstation and a connected controller have different effects. The connected system may cause physical action, so the teacher identifies the permitted practice environment.

Support the explanation

1. Set your intention and keep a sketch reference

Expected evidence identifies the purpose of practising dimensioned, transferable CAD geometry; shows or references a labelled square and central hole; and identifies where the student's source is stored. Its caption explains the dimensions or layout shown. The participation explanation remains separate from any claim of machine competence. A different design is acceptable when explicitly proposed and awaiting the teacher's confirmation.

Look for this evidence

- An intention with a clear learning purpose.
- A sketch or traceable reference with dimensions, central relationship and caption.
- Student preparation and teacher-operated demonstration roles described accurately.

Valid alternatives

A labelled paper sketch is valid; no image upload is necessary.

An alternative proposal may be recorded with teacher confirmation still required.

Original extended response

Explain how a group can contribute to a proposed plasma project and observe a teacher demonstration responsibly. Compare offline preview with SureControl Simul, explain the separation and access boundaries, and distinguish learning evidence from operating authorisation.

Illustrative response

Our group can sketch, dimension, check files and explain choices on an approved offline computer. An offline CAM preview displays a path; SureControl Simul followed by Start moves the real machine and remains teacher-only. Students observe from the approved position outside the work area. During automatic cutting, the trained teacher is also separated by screens and welding curtains. Entry requires the school's electrical isolation procedure and teacher permission; Stop or Offline is insufficient. We can record what we learned or actually observed, without claiming that a quiz or journal authorises operation.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Explains that SureControl Simul causes real movement and remains teacher-only.
- Includes operator screening and student access only after required electrical isolation.
- Keeps learning evidence separate from operating authorisation.

A useful discussion example

A group has checked its DXF on an offline computer. They can record: 'We checked the imported dimensions and explained the proposed path.' If the teacher later demonstrates a controlled machine dry run, the group can record what they actually observed from the approved viewing area. Neither record means the students operated the machine or are authorised to do so. Keep any unobserved or unconfirmed step clearly labelled.

Student evidence page 8. Full-size sketch and selected-image fields: student page 49.

Dimensions and datums

01 B — Which source best communicates the intended size of a particular part?

A dimensioned part drawing with units. Dimensions and units communicate the intended geometry of that part.

02 C — A dimensioned 100 mm line is printed at 1:2. What length does its dimension specify?

100 mm. The dimension describes the intended length regardless of its reduced representation.

03 A — Which check establishes imported size more reliably than screen appearance?

Read the actual dimensions in the receiving application. Numerical checks establish the imported geometry's size.

04 B — What radius produces the proposed Ø30 mm hole?

15 mm. The radius is half the 30 mm diameter.

05 C — A student enters 30 mm into a radius field. What diameter results?

60 mm. Diameter is twice radius, so this entry creates a different hole from the proposed example.

06 A — Where is the centre of a square extending from (0, 0) to (100, 100)?

(50, 50). The midpoint is 50 mm from either bound on both axes.

07 B — Why can the centred Onshape square and the 2D Design square have different coordinates but the same size?

They use different datums while each spans 100 mm. A change of coordinate reference need not change the geometry.

08 C — What does the CAD origin establish by itself?

The zero reference used to describe drawing coordinates. That is its role; relating the job to the machine requires deliberate preparation.

09 A — Which description correctly distinguishes the SureControl labels?

Set Origin assigns job zero; Origin commands movement to it. The distinction matters because one of these labels represents a motion command.

10 B — Why should a supplier screenshot have a clear caption?

It distinguishes a reference example from the current school installation or the student's result. Context prevents a useful reference being mistaken for evidence it does not provide.

Support the explanation

Which drawings describe the same shape?

A: width 100 mm, height 100 mm, diameter 30 mm, central at (50,50). B: width 100 mm, height 100 mm, diameter 30 mm because radius 15 mm is doubled, central at (0,0). A and B are equivalent despite their datums. C has diameter 60 mm because radius is 30 mm, so its hole differs. The annotation distinguishes datum and centre and explains that machine referencing/job placement remains teacher controlled.

Look for this evidence

- Correct spans and conversion of both radius entries.
- A sketch whose labels establish measurements regardless of printed scale.
- Drawing geometry distinguished from physical placement.

Valid alternatives

Either equivalent datum is valid.

A freehand diagram is acceptable with coherent labels; full-size drawing is unnecessary.

Original extended response

Compare the two coordinate layouts for the proposed 100 mm square with a central $\varnothing 30$ mm hole. Explain how you would establish that they represent the same geometry, and identify the information the trained teacher still needs before relating the drawing to a job on the table.

Illustrative response

The first square spans 0–100 mm on both axes, with centre (50,50). The centred square spans –50 to +50 mm, also 100 mm, with centre (0,0). Both have a central $\varnothing 30$ mm hole; its radius is 15 mm. I would inspect actual units, widths, heights and circle dimensions rather than zoom or coordinate numbers alone. These checks establish equivalent drawing geometry. The teacher still needs the intended revision, datum, actual stock position and verified machine reference/job origin before relating either drawing to a job on the table.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Correctly identifies equivalent 100 mm spans and central $\varnothing 30$ mm holes despite different coordinates.
- Explains radius/diameter and the need to inspect actual units and dimensions.
- Keeps physical positioning and machine controls within the teacher's verified procedure.

A useful discussion example

Drawing A runs from $x = 0$ to 100 mm; drawing B runs from $x = -50$ to +50 mm. Each width is 100 mm. Their y spans are also 100 mm. A circle centred halfway across each span is central even though its coordinates differ. Confirming this equivalence checks the drawings. It does not establish where either part lies on the physical table or authorise a machine-origin action.

Student evidence page 11. Full-size sketch and selected-image fields: student page 49.

Draw the practice shape

01 C — Which statement best records design intention for this practice sketch?

It should be 100 mm square with a central Ø30 mm hole. This specifies dimensions and a meaningful relationship to check.

02 A — What does a coincident relationship establish?

The points occupy the same location. Coincidence records the intended shared position.

03 B — How does a driven or reference dimension differ from a driving dimension?

A reference dimension reports a measurement rather than controlling the shape in the same way. The distinction explains why displaying a value does not necessarily establish a design control.

04 C — In the centred Onshape example, what keeps the square and circle aligned?

Anchoring their centres at the same sketch origin. The shared centre establishes alignment while dimensions control size.

05 A — A sketch reports a dimension conflict. What is the appropriate next step?

Inspect existing dimensions and relationships and resolve the conflict. Conflicting information needs correction before accepting the sketch.

06 B — Which inputs create the circle in the documented 2D Design example?

Radius 15 mm; centre (50, 50). These inputs produce the central Ø30 mm circle in the 0–100 square layout.

07 C — Why check Absolute mode before entering the two rectangle corners?

It ensures the positions are interpreted from the drawing origin rather than a previous point. The example relies on absolute positions, so mode changes can alter the resulting geometry.

08 A — Which comparison establishes that the two examples describe the same intended shape?

Both have 100 mm spans and a central Ø30 mm circle. Equal spans and centre relationships establish equivalent geometry despite different datums.

09 B — What should be preserved when exporting a DXF?

The native master and a traceable exported revision. Keeping both allows future changes to be made deliberately and traced.

10 C — A native sketch has passed its checks. What can the student honestly record?

The sketch is checked in CAD; transfer checks remain a separate task. This accurately separates the evidence from each stage.

Support the explanation

2. Record the CAD dimensions and revision

Report actual checked values, not copied expected values. Expected geometry is width/height 100 mm and central diameter 30 mm. Onshape may use shared origin, $-50/+50$ bounds and diameter 30 mm; 2D Design may use absolute bounds 0/100, centre (50,50) and radius 15 mm. Record the master/revision accurately. Incomplete or planned checks are acceptable when labelled, but do not establish a verified CAD result.

Look for this evidence

- Identified editable master or honestly labelled planned record.
- Actual dimensions, units, datum and method, or explicit unchecked status.
- Revision trail linking a change to its reason.

Valid alternatives

Either documented CAD pathway is valid.

Accept “Not checked yet” or a planned drawing rather than invented software results.

Original extended response

Explain how you would establish and check the intended square-and-hole geometry in either Onshape or 2D Design. Compare that method with the other pathway and justify how the editable master and export record preserve the design intention.

Illustrative response

In Onshape I would use millimetres, a 100×100 mm centre rectangle and a $\varnothing 30$ mm circle, both centred on the sketch origin. Dimensions establish size and the shared centre establishes position. I would inspect relationships and resolve conflicts. The equivalent 2D Design drawing uses absolute corners (0,0)/(100,100), centre (50,50) and radius 15 mm. I would preserve the Onshape document/version or DTD master, export a named DXF revision and record choices. Native checks and later receiving-software checks remain separate evidence of the design and its transfer.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Correctly explains dimensions and the method used to locate the centre.
- Compares the two coordinate layouts without treating them as different-sized parts.
- Preserves the native master and separates native checks from transfer checks.

A useful discussion example

A circle appears close to the middle of an Onshape square, but the centre relationship has not been confirmed. The drawing's appearance is insufficient. Inspect the intended coincidence with the origin and the dimensions, then resolve any conflict. In 2D Design, inspect the stated centre coordinates and object properties instead. Keep the resulting native master and its named export so later feedback can be traced to the checked revision.

Student evidence page 14. Full-size sketch and selected-image fields: student page 49.

Geometry that can be cut

01 A — What are the intended contours in the proposed square-and-hole drawing?

One closed outer square and one closed inner circle. These describe the two intended boundaries of the retained plate.

02 C — A square is exported as four lines whose endpoints form a closed loop. How should it be judged?

It can represent one closed contour, subject to the remaining checks. The representation can be legitimate even when its entity count differs.

03 B — Why can two apparently touching endpoints still need inspection?

A small actual gap may be hidden by the display. Geometry relationships and checks can reveal what the picture conceals.

04 A — What is a likely concern with duplicate lines?

Repeated or ambiguous path information. Coincident duplicates can introduce processing problems even when the drawing looks unchanged.

05 C — An outline crosses itself accidentally. What should be investigated?

The intended contour and retained material at the intersection. The design must communicate an unambiguous boundary before production preparation.

06 B — What is the sound approach when a receiving program handles a spline poorly?

Keep the master, make a recorded trial conversion or export change, and inspect the result. A traceable trial lets you check both compatibility and the preserved shape.

07 A — Why might an O-shaped opening need a bridge in a larger plate?

Its retained centre would otherwise be isolated from the surrounding plate. A deliberate connection can retain an island, subject to a suitable reviewed design.

08 C — When would the centre of the same ring shape reasonably be waste?

When the intended product is the separate metal ring itself. In that case the centre may be deliberately removed rather than retained.

09 B — What does successfully opening the supplied unitless logo DXF establish?

That the application could read the file; size and contour checks are still needed. Readability is useful but is a narrower claim than verified geometry or production approval.

10 A — Which is a defensible first step in a repair exercise?

Preserve the original and establish intended size before making a labelled working revision. This keeps the work traceable and avoids guessing a physical scale.

Support the explanation

Find what needs checking in an outline

A meets the contour requirement: several joined entities can form one closed contour. B identifies a gap; C a duplicate line. D lacks reliable size evidence. E needs a retained-material decision and teacher review of a connection if the centre must stay. Preserve the original/master, change an identified copy, correct the join or unwanted duplicate, then export and recheck units, dimensions and contours. A supplies no production compatibility or operating approval.

Look for this evidence

- All five classifications with specific reasons.
- Original-versus-corrected sketch and traceable repair/recheck for B or C.
- Closure alone is not complete file or production readiness.

Valid alternatives

Accept equivalent category wording.

Either gap or duplicate repair is valid; do not require both.

Original extended response

Evaluate a proposed handoff of a logo that opens successfully but has unset units, several complex curves and a letter centre that may become detached. Explain a source-preserving inspection and revision plan, and justify why successful opening is insufficient evidence of readiness.

Illustrative response

I would preserve the logo's original file and inspect a labelled working copy. Unset units mean intended size needs reliable evidence before scaling. I would identify retained regions and check closed contours, gaps, duplicates and self-intersections. A letter centre meant to remain needs a connection reviewed for the actual material and process. Complex curves need a compatibility trial that retains the master and checks shape after re-import. Each revision needs its own record. Successful opening proves only readability; unresolved dimensions or boundaries stop handover, and no file check grants cutting approval.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Uses specific geometry and unit checks rather than visual appearance alone.
- Explains the retained-island issue in terms of design intention.
- Preserves a traceable source and rechecks revised exports.
- Does not infer cutting approval from successful import.

A useful discussion example

A logo imports successfully, yet its intended size is unknown and a letter centre may fall free. Keep the original, label a working copy and seek reliable size evidence before scaling. Identify retained material, inspect contours and review curve compatibility. If a centre must remain attached, propose a connection for teacher review. Re-import the revision to check it. Record unresolved information rather than treating successful opening as a cutting approval.

Student evidence page 17. Full-size sketch and selected-image fields: student page 49.

Design with retained material

01 B — Which starting point gives the clearest basis for a design decision?

State who will use the product and what it should help them do. A clear purpose provides reasons for decisions and later evaluation.

02 A — How should the equipment-label example be understood?

It is an example for design discussion, not approval to manufacture it. The lesson uses the concept to explain design reasoning.

03 C — Which criterion is most directly checkable?

The intended viewer can recognise the message from the agreed viewing position. The designer can test this outcome with an appropriate review.

04 B — Why might two useful criteria conflict?

Extra visual detail may strengthen identity but create difficult small features. The designer must explain and review this trade-off rather than favouring detail automatically.

05 A — What does feasibility concern here?

Whether the proposal can be achieved within its actual requirements and constraints. Feasibility considers the approved material, process and relevant limits as well as the drawing.

06 C — A design includes a narrow connection of uncertain suitability. What is the appropriate conclusion?

It needs review against the confirmed material and process requirements. Suitability depends on the real constraints and the trained teacher's review.

07 B — Which action is a useful iteration?

Revise the lettering after a readability review, preserve the earlier version and check again. The revision addresses a specific issue and creates evidence about its effect.

08 A — Why use paper or on-screen review before cutting a concept?

It can reveal design issues early and reduce avoidable trial waste. Digital or paper checks can support revision before consuming metal.

09 C — Which claim is supported by a CAD measurement record alone?

The stated drawing dimensions were checked in CAD. This accurately identifies the stage and result of the check.

10 B — What makes a final improvement proposal useful?

It identifies a specific change, why it is needed and how it will be evaluated. This keeps the next iteration purposeful and testable.

Support the explanation

Choose a revision for a clear reason

B is the better-supported starting point: the viewer recognises it without prompting and it proposes connections. Those connections remain untested against actual material/process requirements. A useful next check could confirm connections with the teacher or test a readability revision with the intended group. Preserve the earlier design and identify the revised master/export. Choosing A is defensible only if the reasoning addresses both crowded lettering and detached centres through a checkable plan.

Look for this evidence

- Two comparison rows separating supplied findings from missing evidence.
- Development choice tied to the group recognising its gear.
- One specific change/check, success evidence and revision trail.

Valid alternatives

Either concept is valid with a defensible plan; B has stronger supplied evidence.

Valid revisions include spacing, recognition or retained-connection review without invented size rules.

Original extended response

Justify a revision to the proposed equipment-label concept using its intended user, purpose and two checkable criteria. Explain the evidence you would collect, the requirements that need confirmation before production, and how you would avoid claiming a physical result from digital evidence alone.

Illustrative response

The proposed label helps the intended group recognise its gear. I would compare recognition from the agreed viewing position and connected retained areas. B has stronger supplied recognition evidence, so I would develop it while asking the teacher to review its proposed connections. A labelled viewer trial and geometry inspection can test the revision. I would preserve the earlier version, change the master and recheck its export. Material, thickness and process limits still need confirmation. This supports a design decision and digital checks, not a claim that a physical label has been successfully cut.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Connects the proposed revision to purpose and observable criteria.
- Identifies a realistic evidence method and relevant constraints.
- Preserves traceable iteration and distinguishes digital from physical evidence.
- Keeps manufacturing approval and operation within the teacher's process.

A useful discussion example

A proposed label is recognisable in a large sketch, but its detailed lettering includes disconnected centres. The student compares a simpler layout against the same recognition and connectivity criteria. They revise the lettering in the editable master, retain the first version and identify a new export. The record explains which centres are now connected and seeks teacher review of the remaining narrow features. This supports a design improvement; it does not claim a successful physical cut.

Student evidence page 20. Full-size sketch and selected-image fields: student page 49.

Check the DXF handover

01 A — Why retain the native CAD master after exporting DXF?

It preserves the editable design information. The master supports later dimensional and geometry changes; DXF is a checked exchange copy.

02 B — Which record is the editable master in the Onshape pathway?

The Onshape document and named version. These retain the editable design and a clear revision.

03 C — What should the documented Onshape export include?

The whole intended sketch. The complete sketch carries the intended outer geometry and hole.

04 A — Which 2D Design route creates the documented exchange copy?

File > Export File, then DXF File. That is the DXF export route confirmed in the installed help.

05 B — Why is a drawing that fits the window not yet size-checked?

Zooming or fitting can hide a size difference. The drawing must be measured in its interpreted units.

06 C — The imported circle reports a radius of 15 mm. What does this mean for the 30 mm diameter practice hole?

The radius agrees with the intended diameter. Diameter is twice radius, so these labels describe the same hole size.

07 A — Which import choice needs particular care when preserving intended size?

Fit to media. The documented option can rescale a drawing; use deliberate units and scale.

08 B — Two squares have bounds -50 to +50 and 0 to 100 mm. What can be concluded?

Both span 100 mm despite different coordinate origins. The spans agree; the origin labels alone do not show a size error.

09 C — A master is v2 but its exchange copy is v1. What should happen?

Export and check an identified copy of the revised master. The exchange file must correspond to the intended design revision.

10 A — Which handoff statement is most useful?

The revision, units, export choices and completed checks are recorded. These details make the exchange traceable and reveal remaining checks.

Support the explanation

Does the drawing keep its size?

Both coordinate layouts span 100 mm each way with a central Ø30 mm hole; R15 mm is equivalent. A paper comparison cannot claim a software round-trip. For an actual import, expect the file/revision, application, unit/scale choice and measured properties, including centrality and extra geometry. Fit-to-window appearance is not measurement. A mismatch leads back to preparation/master correction and an identified re-export rather than cutting-setting changes.

The student table separates the centred and corner-origin layouts. In each row, distinguish what the written geometry supports from checks still needed after an actual DXF import. These are not additional A/B/C sample records.

Look for this evidence

- Expected, calculated or measured values distinguished with units and feature labels.
- Source/file revision and check method identified.
- Next action follows the result; untested behaviour remains explicit.

Valid alternatives

A labelled paper comparison meets the source alternative.

Actual offline import evidence is valid when software is provided and the student records real checks.

Original extended response

Explain how you would check and hand over a revised training-square DXF when its coordinate origin differs from the example. Justify the checks that matter and state what your evidence cannot prove.

Illustrative response

I would identify the revised master and its matching DXF, reopen the export separately and inspect millimetres, scale, overall dimensions, circle size and complete contours. A span of -50 to +50 mm equals 0 to 100 mm, so a different datum need not mean a size error. Ø30 mm also equals R15 mm. I would record the actual application, revision and measurements, checking stray geometry. A failed check returns to the master and a new export. These checks establish transferred digital geometry, not physical placement, toolpath suitability or machine approval.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Uses dimensions and units to justify the comparison rather than relying on appearance.
- Keeps the revision traceable and distinguishes CAD evidence from machine approval.

A useful discussion example

The master records a hole diameter of 30 mm. After export, the receiving program reports a radius of 15 mm. These values describe the same circle because the diameter is twice the radius. The student then checks the imported width and height, confirms the hole is central and records the units and revision. Matching the circle alone does not establish the rest of the drawing, and these digital checks do not prove a physical cutting result.

Student evidence page 23. Full-size sketch and selected-image fields: student page 49.

From outline to toolpath

01 B — What does CAM add to a checked CAD outline?

A planned toolpath, operation and cutting sequence. CAM combines geometry with an operation and movements; CAD geometry alone does not contain that complete plan.

02 C — A DXF passed a CAD round-trip. What should happen after import into SheetCam?

Check units, dimensions and geometry again. Each receiving program can interpret the transfer differently, so the CAM import needs its own check.

03 A — Why inspect the contents of a selected layer?

The layer may include unintended or duplicate geometry. The operation acts on selected entities; its layer name cannot validate them.

04 B — Why is the torch path offset from a finished contour?

The kerf removes material on both sides of the path. Offsetting allows for the removed strip so the intended material can remain.

05 C — For the practice square and hole, where is the waste relative to each contour?

Outside the square and inside the hole. The retained material lies inside the outside profile but outside the opening.

06 A — The operation is named Outside Offset. What is the best review?

Inspect the generated path around every closed contour. The supplier example can interpret outer and inner contours; the generated path still needs review.

07 B — Why can a part fit on the sheet while its planned operation still needs review?

Lead-ins, lead-outs and travel can extend beyond the outline. The planned route can occupy space outside the nominal part footprint.

08 C — Why are internal features normally cut before the outside profile?

The surrounding material still supports the part. Keeping the outside profile attached helps retain support while internal features are made.

09 A — Which statement correctly distinguishes the two simulations?

SheetCam previews offline; SureControl Simul can move the real machine. These similarly named features have different physical consequences; machine movement is teacher-only.

10 B — Why is a .tap filename insufficient to confirm compatibility?

The postprocessor and installed system must also match. The extension names a file type, while the selected postprocessor and installation determine suitability.

Support the explanation

Trace the file handoff

Order: editable CAD master → checked DXF → editable CAM job with reviewed toolpaths → machine-specific TAP. The master preserves design; DXF transfers geometry; CAM retains operations/route; the teacher's verified post-processor produces compatible output. The v1 DXF is stale when v2 is intended. A correction names the v2 master and matching re-export, repeats checks and marks v1 superseded. Timestamps or renaming do not repair content. Pending output compatibility stays with the teacher.

Look for this evidence

- Four ordered records with distinct purposes and next conversion/check.
- Specific revision correction, not “use the newest file”.
- Handover separates offline work from teacher verification without generating code.

Valid alternatives

Either Onshape document/version or 2D Design master is valid.

CAM review can remain a labelled written proposed review when software is unavailable.

Original extended response

Explain how you would review a CAM preview of the practice square and hole before handing it to the authorised teacher. Justify why a correct-looking outline and successful preview are insufficient for production approval.

Illustrative response

I would compare imported units, dimensions, hole position and selected geometry with the identified master. Kerf removes material, so I would inspect the path on the correct waste side of each boundary: outside the outer profile and inside the hole. I would follow internal-feature-first sequence, lead space, start points and rapid moves. An offline preview supports route inspection but cannot test physical clearance or cutting conditions. The authorised teacher must verify the material/process, post-processor and installed-system compatibility. A TAP extension or successful preview alone does not establish production readiness.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Connects geometry, offsets and sequence to the intended part.
- Distinguishes offline evidence from physical-machine and compatibility approval.

A useful discussion example

For the practice square, the metal to keep lies inside the outside boundary but outside the circular hole. The planned outside path therefore lies in external waste, while the hole path lies within the opening's waste. The student follows each contour in the offline preview, checks leads and travel, and confirms that internal features are planned before the outside profile. These checks explain the route; they supply neither a numerical kerf value nor permission to run the machine.

Student evidence page 26. Full-size sketch and selected-image fields: student page 49.

A traceable file handover

01 B — What is the main purpose of nesting?

Arrange the required parts while allowing a workable route. A useful nest considers material use together with cutting paths and movements.

02 C — What should happen when a part is copied in a nest?

Check that the added instance matches the required quantity. A copy adds another part instance that must be included in the job record.

03 A — A part is too large for the available space. Why not simply shrink it?

Scaling changes its dimensions and needs an intentional design decision. An arrangement problem must not silently change the specified part.

04 B — Two outlines fit side by side. What still needs inspection?

Leads, offsets, travel and spacing in the complete route. The planned movements can occupy space beyond the finished shapes.

05 C — Which spacing value should students assume for every nest?

No universal value; use the confirmed teacher-reviewed process. Required spacing depends on the actual route and process, so a generic gap is not established here.

06 A — Which record retains arrangement and operation work for revision?

The CAM job. The CAM work retains the arrangement and operation decisions behind the output.

07 B — A hole is moved in the CAD master. What follows?

Review and update affected exports, CAM work and output. Downstream records based on the earlier geometry may now be stale.

08 C — Which handoff label is most useful?

A project, revision and stage label consistent with its contents. The label helps identify the intended file when it agrees with the handoff record.

09 A — What can a student honestly record after reviewing only an offline preview?

The offline preview was reviewed, with observations listed. This records the scope of real evidence without claiming a physical test.

10 B — A tidy nest has an uncertain revision. What is the useful next step?

Resolve which source and output revision belong together. The actual contents must match before the handoff can be trusted.

Support the explanation

3. Record the DXF handoff checks

Name the current DXF and matching master, actual export choices and millimetres. Identify the receiving application, actual width/height/hole measurements, contours, extra/missing geometry and unresolved issues. For the proposed nest, request quantity, route/lead-space and revision evidence. If no import occurred, “Not tested yet” and a concrete next action are preferable to invented results. A filename or TAP extension alone cannot establish compatibility or readiness.

Look for this evidence

- Traceable master/DXF with actual choices and units.
- Completed receiving checks distinguished from pending checks.
- Status follows evidence; tidy layout is not production approval.

Valid alternatives

Incomplete handover is valid formative work with explicit gaps and next checks.

A teacher-approved naming convention is valid without copying the example filename.

Original extended response

Evaluate a proposed handoff containing a tidy nest, two differently dated DXFs and a file named ready-to-cut.tap, but no revision record. Explain how you would make the handoff useful without claiming machine approval.

Illustrative response

The tidy nest shows an arrangement, not sufficient lead space, correct quantity or an approved route. I would identify the intended master and revision, trace each DXF to it and mark stale files superseded. I would record units, dimensions, quantity, rotations, overlaps, leads and path sequence, then identify the related CAM work and output. File dates and ready-to-cut.tap cannot supply this missing history. The authorised teacher still verifies actual stock, process, post-processor and installed compatibility. I would pause the handover until the revision is clear without claiming any machine test.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Justifies a clear relationship between design revision, exchange geometry, CAM work and output.
- Distinguishes completed offline checks from production compatibility and authorisation.

A useful discussion example

A student moves the hole in practice-square-v02 but hands over the v01 DXF because its filename still looks familiar. The new master and old exchange file no longer describe the same design. The student preserves the earlier records, makes a matching export and repeats geometry checks. If CAM or output already exists, those stages also need review and regeneration as appropriate. The handover states the completed digital checks and leaves production compatibility and operation to the teacher.

Student evidence page 29. Full-size sketch and selected-image fields: student page 49.

Read the controller screen

01 B — What is the first useful question when reading an unfamiliar screen item?

Does it display information, report state, select an option or command an action? Classifying the item helps prevent confusing a report with a control that changes the machine.

02 C — A familiar job outline is visible. What does that establish?

A job is displayed, while revision, size and physical setup still need checking. A familiar picture is only one piece of information.

03 A — Why read the unit beside a feed-rate override value?

An override percentage is not a feed in millimetres per second. The percentage modifies a feed value rather than describing distance travelled per second.

04 B — The controller says Idle. Which interpretation is justified?

It reports a controller state, not electrical isolation. Idle does not establish that the required electrical isolation has occurred.

05 C — A file is loaded but the controller requires an origin. What does this show?

Loading the file does not complete every preparation stage. The teacher must resolve the required state within the approved sequence.

06 A — How does an input indicator differ from a selectable option?

An indicator reports a signal; an option enables or selects a function. They have different roles even when their labels resemble one another.

07 B — The Touch option is selected. Which evidence is still needed to verify the sensing function?

Whether the expected sensing signal operates during the authorised check. The selected option alone does not test the contact signal.

08 C — What is the difference between Set Origin and Origin?

Set Origin assigns job zero; Origin commands movement to it. The similar labels have different consequences, so both belong to the authorised teacher sequence.

09 A — What happens when SureControl Simul is started?

The real machine moves without torch firing. It is a physical dry run and remains teacher-only.

10 B — Where must the operator be during automatic cutting?

Physically separated by the securely screened operating area and welding curtains. The supplied school requirements apply to the operator as well as students and bystanders.

Support the explanation

Review the proposed working arrangement

A disagrees: Simul followed by Start moves the real machine, so a trained, authorised teacher operates it under controlled-area arrangements. B disagrees: the operator must also be separated by the screened area and welding curtains during automatic cutting. C agrees with offline preparation; it does not authorise operation. A valid unresolved fact is current operator training, isolation or actual screening, confirmed through school training/approval/site procedure rather than a screenshot.

Look for this evidence

- A/B rejected with movement/separation reasons; C accepted only within offline scope.
- Clear corrections linked to guide stages or workbook lessons.
- One local fact and appropriate confirmation evidence.

Valid alternatives

Accept stage title or workbook lesson as a source reference.

Several local facts are valid; require one concrete fact and credible confirmation route.

Original extended response

Explain why a student observing a loaded job, Idle status and a selected Touch option cannot conclude that the machine is ready to cut. Use the difference between information, signals, options and actions.

Illustrative response

The job picture shows loaded geometry, but does not confirm revision, size, sheet placement or the actual material. Idle describes a controller state and does not prove electrical isolation. A selected Touch option enables a function; the Touch input indicator reports a sensing signal, which still needs the authorised teacher's check. Origin commands movement, while Set Origin assigns the current location as job zero. Students interpret these differences without clicking. The teacher completes physical and compatibility checks, and remains physically separated by screens and welding curtains during automatic cutting.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Uses accurate distinctions among display, controller state, input indicator, option and action.
- Explains why screen observations cannot replace physical checks, authority or required separation.

A useful discussion example

A supplier screen shows a familiar square, an Online option and an Idle state. A student says, 'It is ready to cut.' The stronger response identifies only the visible evidence: a job is displayed, an option is selected and the controller reports Idle. Revision, dimensions, origin, material and physical arrangements require separate checks. Idle is not electrical isolation, and none of these observations gives a student permission to start, jog, reference or simulate machine movement.

Student evidence page 32. Full-size sketch and selected-image fields: student page 49.

Understand the settings

01 B — What makes a cut-chart row relevant to a job?

Its material, thickness and specified setup match the actual process. The chart row is meaningful only when the identified equipment and process conditions match.

02 C — How should a supplier screenshot's cutting values be used in this course?

To understand the fields, without assuming a school recipe. The screenshot explains software; the actual recipe still needs a matched authoritative chart.

03 A — Why retain the unit when recording feed rate?

The numerical expression changes with the time unit. Millimetres per second and per minute use different numbers for the same physical feed.

04 B — Which quantity is expressed as a percentage in this explanation?

Feed-rate override. The override expresses a relative change to the base feed.

05 C — What happens to physical amperage when the software Amp field is changed?

The software field does not change the physical current setting. The manual requires the physical source control to be checked separately.

06 A — Software Amp matches the chart, but the source control has not been checked. What is missing?

The authorised teacher's physical current check. A matching software field cannot establish the actual source setting.

07 B — Why are pierce height, pierce delay and cut height separate fields?

They describe different positions or timing stages. Position for piercing, time allowed for piercing and cutting position have distinct purposes.

08 C — What can make a recorded base feed misleading?

An unnoticed feed-rate override. The override can change the commanded feed relative to its recorded base value.

09 A — What does THC use in the documented control process?

Arc-voltage information to adjust torch height. The target voltage belongs to the height-control process and is different from physical amperage.

10 B — A cut edge is rough. What conclusion is justified?

Several causes are possible; record evidence and use the approved review pathway. Speed, height, consumables and alignment can interact, so the symptom does not identify one certain fix.

Support the explanation

Read the label before the number

Amp selects software process information; it does not change physical power-source amperage. Feed is distance/time: 1 mm/sec equals 60 mm/min, so copying a bare number may create a sixty-fold mismatch. Pierce height positions the torch for piercing, shown in millimetres in the source metric setup; it is distinct from cut height and pierce delay. Values require the teacher's matching chart, equipment, torch/consumable and material/thickness record. An unlabelled speed screenshot establishes no unit or approved value.

Look for this evidence

- Three distinct meanings, units/uncertainty and approved-source requirements.
- Correct unit relationship and physical/software current distinction.
- Screenshot limitation identified without invented values.

Valid alternatives

Accept mm/s or mm/min notation when explained correctly.

An actual-installation unit may remain explicitly unconfirmed.

Original extended response

Evaluate a settings record that includes material thickness and software Amp but omits the consumable identity, feed unit and physical current check. Explain why completing those gaps matters before the teacher considers production.

Illustrative response

Material thickness alone does not identify a matching chart row. The teacher must confirm material, power source, torch and consumable identity against the approved process information. Missing consumable identity may mean the row describes a different setup. Feed without a unit is ambiguous because 1 mm/sec equals 60 mm/min. Software Amp selects software information and does not set physical current; the power-source control needs a separate check. I would request those records and the actual current verification, preserving gaps until resolved. I would neither invent replacement values nor enter settings into the controller.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Connects each missing item to a specific process or interpretation problem.
- Proposes evidence-based checks within the teacher-only production boundary, without inventing a recipe.

A useful discussion example

A chart expresses feed in millimetres per minute, while a supplier software example labels feed in millimetres per second. Compare the quantities first. Divide the per-minute value by sixty to express the same feed per second, then check the documented units and override. This arithmetic checks consistency; it does not choose a cutting recipe. The teacher must still match equipment, material and process, and verify physical amperage separately from the software Amp field.

Student evidence page 35. Full-size sketch and selected-image fields: student page 49.

Observe the production sequence

01 B — What does completing this course's formative work establish?

Evidence of learning, not machine-operator authorisation. Understanding supports preparation but does not replace the required training and approval.

02 C — Why match the design revision, production output and actual process?

The correct file and correct process must also belong together. A stale file or mismatched material setup can invalidate the intended job.

03 A — Which statement includes the operator-separation requirement?

The operator is physically separated by the securely screened area and welding curtains during automatic cutting. The separation requirement applies even to the authorised teacher.

04 B — When may students enter the shielded area?

Only after the required electrical isolation and within teacher directions. The supplied school requirement does not permit access merely because motion has stopped.

05 C — What does reference establish in the documented sequence?

A known machine position through real axis movement. Reference is a physical movement stage, not a file-opening action.

06 A — An expected Touch signal does not appear during a teacher check. What is the student's role?

Observe and report while the teacher follows the stop and support pathway. Students do not repair sensing, change configuration or continue the sequence.

07 B — Why does SureControl Simul require a controlled operating area?

It moves the real machine when started. Removing torch firing does not remove the motion hazard or teacher-only boundary.

08 C — The offline preview was correct but the physical route would exceed the usable sheet. What follows?

Stop and resolve the setup through the approved review pathway. Correct design dimensions do not prove that the physical placement is suitable.

09 A — A fault occurs during production. What should students do?

Follow directions and report observations while the teacher applies the approved response. Students do not approach, restart or change settings during the fault.

10 B — Why is a motionless finished part not immediately ready to retrieve?

Required isolation, cooling and handling checks still apply. Stopped movement does not remove electrical, heat or sharp-edge concerns.

Support the explanation

4. Keep an observation or feedback record

Identify genuine evidence and provenance: direct observation, teacher-reported feedback or a permitted sample. Connect it to a revision/sample and describe a specific finding plus limit. If only the printed scenario is available, label it hypothetical rather than witnessed machine use. “Awaiting observation or feedback” is appropriate when none occurred. Student operation, uncontrolled retrieval and invented trial outcomes cannot be valid evidence.

Look for this evidence

- Evidence type and revision/sample context identified.
- Observation distinguished from reported feedback.
- Claim remains within actual events, including absence of observation.

Valid alternatives

Teacher feedback on an offline drawing is fully valid; no live cut is required.

A pending entry can honestly record what remains to be done.

Original extended response

Explain how a class can learn from a teacher-controlled demonstration when the offline preview is correct but the physical dry run fails the fit check. Justify the pause, the student role and the conditions that still apply before any later cut or retrieval.

Illustrative response

The offline preview checks the proposed route, but cannot prove its placement on the real sheet. A failed physical fit check means the trained teacher pauses, follows the approved review pathway and resolves placement or setup before repeating required checks. Students stay clear and record the observed mismatch without changing controls. Any later automatic cut still requires authorised operation and physical separation through screens and welding curtains. Stopping or selecting Offline does not establish electrical isolation. Retrieval waits for the teacher’s isolation, cooling and handling procedure; a quiet part may remain hot or sharp.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Explains the purpose and limits of each kind of check rather than treating the pause as failure.
- Maintains teacher-only authority, screened operator separation and the required isolation and handling boundary.

A useful discussion example

An offline preview shows the intended part, but a teacher-controlled physical dry run would travel outside usable sheet space. The student records the observed placement problem without claiming the CAD dimensions are wrong. The teacher stops and reviews the setup through the approved pathway before repeating required checks. No live cut is used to test the uncertainty. Student evidence distinguishes the offline review, authorised physical observation and correction that still needs verification.

Student evidence page 38. Full-size sketch and selected-image fields: student page 49.

Measure before judging

01 B — When can a student begin inspecting the physical sample?

After the teacher releases a safely handled sample under the approved procedure. Teacher clearance connects isolation, cooling, retrieval and the permitted inspection method.

02 A — Why record a sample's file revision and recipe identifier?

They connect the observed result to the job that produced it. A traceable record allows the operator to compare the actual program and recipe with the sample.

03 C — A drawing specifies a 30 mm hole diameter. Which measurement compares the same feature?

The hole diameter measured in millimetres. Comparing diameter with diameter in the same unit gives a meaningful comparison.

04 B — A proposed 100 mm width measures 99.6 mm. What is measured minus intended?

-0.4 mm, meaning below the intended width. 99.6 minus 100 is -0.4 mm for this measured location.

05 A — A part differs from its drawing by 0.4 mm, but no tolerance is given. What is justified?

Report the difference and discuss the required fit or purpose. The difference is established; acceptability needs an agreed criterion.

06 C — An outside edge has a smaller top width than bottom width. Which description matches the guidance?

Positive bevel. Positive bevel describes an outside part that is smaller at the top than bottom.

07 B — What does a clear photograph of an irregularly sloping edge establish?

The visible edge pattern needs recording and trained review. The image supports an observation, while diagnosis requires more evidence.

08 A — What is useful to record about dross before any approved removal?

Its location and appearance, using permitted photographs or teacher observations. Location and appearance help describe the result without requiring unsafe scraping.

09 C — Why label a width measurement as 'top' or 'bottom' when bevel is present?

Because the measured width can differ between those locations. Bevel can change the top and bottom dimensions, so the location matters.

10 B — Which evidence set best supports a judgement about a cut part?

Labelled measurements plus permitted top, underside and edge observations. These complementary records cover dimensions and several aspects of edge quality.

Support the explanation

Make the measurement report useful

Top width: $99.6 - 100 = -0.4$ mm, 0.4 mm below intended. Hole diameter: $30.3 - 30 = +0.3$ mm, 0.3 mm above intended. Underside deposits are observations; no tolerance or cause is supplied. “Failed” and “speed is wrong” are unsupported. A useful report labels the hypothetical example, retains both measurements and underside location, requests tolerance/process review and prescribes no setting. Real measurement requires a safely teacher-released, identified sample and permitted method.

Look for this evidence

- Correct calculations, signs, features, locations and units.
- Dimensional differences/deposits distinguished from pass/fail and diagnosis.
- Real access, isolation/cooling/retrieval and measuring method remain teacher controlled.

Valid alternatives

Accept “0.4 mm undersize at the top” with explicit calculation/reference.

Equivalent report wording is valid when measurements and limitations remain clear.

Original extended response

Evaluate the evidence you would need to judge a teacher-cleared cut sample. Use the proposed 100 mm square with a 30 mm hole, or your teacher-approved design. Explain what you can conclude from measurements and photographs and what remains uncertain.

Illustrative response

I would begin with a safely teacher-released sample, identify its revision/material and label top and bottom. I would record instrument, method, feature, location and units, then repeat permitted measurements consistently. In the written example, $99.6 - 100 = -0.4$ mm for top width, while hole diameter differs by +0.3 mm. Without an agreed tolerance these are differences, not automatic failure. Labelled photographs can show edge slope and underside deposits, but cannot establish an exact bevel angle or cause alone. The trained operator needs the sample and process record for further judgement.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Connects at least two quality judgements to relevant, labelled evidence.
- Distinguishes measured differences, agreed tolerance and possible causes without inventing an approval or setting.

A useful discussion example

A hypothetical teacher-cleared practice sample has a top outside width of 99.6 mm and a hole diameter of 30.3 mm. The nominal values are 100 mm and 30 mm. Measured minus intended gives $99.6 - 100 = -0.4$ mm for width and $30.3 - 30 = +0.3$ mm for the hole. The signs show which measurement is below or above nominal. Record instrument and locations; without an agreed tolerance, neither difference establishes pass or fail.

Student evidence page 41. Full-size sketch and selected-image fields: student page 49.

Investigate with evidence

01 C — What comes first when production behaves unexpectedly?

The trained teacher applies the relevant safety procedure. Making safe precedes investigation and protects everyone around the equipment.

02 A — Which statement is an observation?

The arc stopped halfway around the outline. This describes the reported event without claiming its cause.

03 B — Why copy the exact fault message and record when it appeared?

Different messages and timings help distinguish possible causes. Specific evidence makes trained diagnosis more focused than a general description.

04 C — An arc-loss report appears. Which conclusion is supported by that report alone?

The interruption needs trained review using the actual evidence. The report establishes an event requiring diagnosis, rather than one definite cause.

05 A — The master is 100 mm wide but the imported DXF is 10 mm wide. What is the next useful check?

Review the file-stage units and export/import records offline. The discrepancy already exists before cutting, so file preparation is the relevant place to investigate.

06 B — What does a successful offline preview establish?

The available geometry and path sequence can be inspected. Offline preview supports file and sequence checks within its scope.

07 C — Why does the supplied guidance consider compensation after other cutting conditions?

Other cutting conditions affect quality and kerf, so sizing changes need context. The order avoids trying to conceal an unresolved cutting problem with compensation.

08 A — Which report is most helpful to the trained operator?

Revision, labelled measurements, observed deposits and the checks still needed. These details separate known evidence from unresolved diagnosis.

09 B — No physical comparison has been performed. How should an investigation be described?

A proposed investigation with the completed checks identified. This reports genuine progress without inventing a physical result.

10 C — The CAD and DXF dimensions agree, but the sample differs. What can a student do?

Preserve the sample and job record for trained review. This keeps the evidence useful while the authorised operator assesses physical causes.

Support the explanation

Choose evidence that supports the claim

A and B are essential: v2 master establishes intended geometry; its matching imported DXF check establishes retained digital dimensions. C needs revision identification. D may be a credited reference comparison, never the student's result. E may show earlier development if labelled superseded. Captions identify v2 and distinguish CAD intention from receiving-software measurements. Physical-size claims need actual teacher-controlled production evidence and permitted inspection, not a supplier photograph.

Look for this evidence

- A/B selected with distinct contributions and matched revision explained.
- Two captions identify digital evidence and revision.
- Evaluation excludes unperformed testing; supporting images/versions are honestly labelled.

Valid alternatives

D/E may support the account with clear limits, but neither replaces A/B.

C becomes useful only after revision and content are established.

Original extended response

Explain a justified next step in this proposed investigation: the master and imported DXF have matching dimensions, but a teacher-cleared sample is undersize and has underside deposits. Use the evidence without prescribing a machine-setting change.

Illustrative response

Matching dimensions in the identified master and imported DXF reduce suspicion of a scale error at that transfer stage; they do not identify the physical cause. I would preserve the sample/revision, labelled measurements, underside-deposit observations, CAM/production records and teacher-recorded recipe for authorised review. Possible speed, height or consumable issues remain hypotheses. Changing several settings would make the effect of each unclear; enlarging CAD could hide the unresolved process difference. The teacher chooses any approved comparison and records its conditions. Until then, my record describes a proposed investigation, not a solved problem.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Uses the matching file dimensions to narrow the investigation without claiming a definite physical cause.
- Proposes traceable evidence and an authorised review process, with honest limits on unperformed tests.

A useful discussion example

Instead of 'bad cut; slow it down', report: 'Hypothetical Sample A, revision v2: top outside width 99.6 mm against 100 mm intended; hole diameter 30.3 mm against 30 mm intended; attached underside deposits. File-stage dimensions matched the drawing. Deposit removal has not been assessed.' This records differences of -0.4 mm and +0.3 mm without declaring acceptance or prescribing a remedy. Retain the sample and recipe record for trained review, and label any future comparison as proposed until completed.

Student evidence page 44. Full-size sketch and selected-image fields: student page 49.

Explain the finished design

01 B — What makes an evaluation stronger than ‘I like it’?

A judgement linked to purpose, agreed criteria and evidence. That connection makes the conclusion understandable and assessable.

02 A — Why distinguish design quality from production quality?

A good drawing can have a poor cut, and a clean cut can reproduce a poor design. The two stages answer different questions and need appropriate evidence.

03 C — Which claim is supported by a master and imported DXF both measuring 100 mm wide?

The checked file transfer preserved that intended width. The records support the digital-scale claim for that feature.

04 B — No physical cut has occurred. What should the evaluation say?

Evaluate the completed digital preparation and identify physical checks still required. This gives a useful evaluation without inventing production evidence.

05 A — A geometry improvement is agreed. What preserves a reliable revision trail?

Revise the master, retain the earlier version, export again and repeat checks. This connects the changed design to a new checked exchange file.

06 C — What trade-off belongs in a proposal to add a bridge to a stencil letter?

The connection can retain the centre but changes the letter’s appearance. That explains both the functional benefit and the visible design consequence.

07 B — A hypothetical 100 mm part measures 99.6 mm and has deposits. What is a balanced conclusion?

Report the difference and edge observations, then seek trained review against agreed criteria. The conclusion uses the evidence without inventing a tolerance or remedy.

08 A — Which caption is appropriate for a supplier bevel photograph in a folio?

Supplier reference example used to compare the edge pattern. This credits the source and states the image’s comparison purpose.

09 C — Which image is most useful beside an observation about underside deposits?

A permitted, labelled underside photograph of the identified sample. That view directly supports the observation and connects it to the sample.

10 B — What should the final presentation make clear about participation?

What the student completed, what the trained teacher operated and what remains proposed. Clear attribution and status make the evidence truthful and easy to follow.

Support the explanation

5. Evaluate the evidence and choose one improvement

Name a criterion and cite a specific earlier record. Evaluation may establish checked digital dimensions or an unresolved transfer issue without any cut. Choose an evidenced improvement, such as a clearer revision record, corrected geometry or better-labelled evidence; explain how a new check tests it. Geometry changes return to the master and new export. Physical quality and operating settings remain with the teacher. “Nothing cut yet” is a legitimate limit.

Look for this evidence

- Criterion-to-evidence judgement with clear scope/limit.
- Specific improvement and practical effect check.
- Traceable revision/contribution statement separating student/teacher roles.

Valid alternatives

Fully digital evaluation is valid.

Improvement may address communication/file preparation rather than alter a correct design.

A proposed bridge can be discussed with appearance trade-off and material/process review explicit.

Original extended response

Justify your final evaluation and one practical improvement to your design or file-preparation process. Connect your judgement to evidence and identify what has actually been completed, what the trained teacher undertook and what still needs checking.

Illustrative response

My criterion was to preserve the practice drawing’s digital dimensions through export. The identified v2 master and receiving-software record both show 100 mm width/height and a central Ø30 mm hole, supporting that limited judgement. I would improve the handover by adding the check method and clearer revision captions, then reopen the package to verify consistency. This improves traceability but needs careful relabelling of older evidence. I prepared and checked the digital files; no physical trial is claimed. Any future production and measurement evidence must come through the trained teacher’s approved process.

Illustrative reasoning, not a claim of student work or a physical trial. Accept equivalent source-supported responses meeting the exact source criteria.

Source success criteria

- Justifies a judgement using relevant evidence while acknowledging its scope and limits.
- Explains a specific improvement and a traceable checking process without inventing a physical result or operational permission.

A useful discussion example

No physical cut has occurred. The student can show an identified master, matching imported dimensions and a reviewed offline path. They conclude that the checked transfer preserved the geometry, then explain a proposed lettering revision and its recognition–connectivity trade-off. They retain the previous version and plan new export checks. The evaluation states that physical dimensions, edge quality and the suitability of bridge sizes still require teacher review; it does not borrow a supplier photograph as proof of personal production.

Student evidence page 47. Full-size sketch and selected-image fields: student page 49.

Close the evidence record

Coordinate reference — page 48

Both illustrated records span 100 mm on each axis and describe a central 30 mm diameter hole; R15 mm equals $\varnothing 30$ mm. The zero point changes the coordinate values, not the shape. X and Y identify the two drawing axes. A CAD datum does not reference the machine or place the job on the physical stock.

Design decision and revision — page 49

Look for two legible selected images or sketches, each identified by file, version and source. The first should establish an original intention or observed problem. The second should show a distinct revision or proposed change. Captions must identify what changed, why, and the next check. Accept paper sketches, CAD prints or references to teacher-approved storage; no image upload is required.

This is student evidence. Keep teacher feedback, practical authorisation and any formal assessment decision visibly separate. “Not checked yet” is appropriate when a check has not occurred. Do not reward fabricated measurements or supplier imagery presented as student work.

Key words and sources — page 50

Use the glossary to correct reasoning rather than test definitions in isolation. Ask students to explain, for example, why two contours need not mean two entities, or why a measured difference is not an agreed tolerance. Reopen the linked source lesson if an explanation conflicts with the compact glossary.

Handover check

Can the reader connect the intended design, native revision, DXF revision, receiving-software checks, teacher production record if any, and the evaluation? Are units and image locations explicit? Are actual, hypothetical and proposed results distinguished? Is the next action specific and within the student’s permitted role?

The source assessment remains provisional and formative. Apply only the formal conditions approved by the school. No student workbook response, progress tick or file submission certifies machine-operating competence.

Review the current assessment and teacher resources